Advanced Digital Signal Processing Coursework

Poor effort. Some parts of the course work are missing. Analysis of the obtained results is often time vague and hard to follow. The report is badly formatted. No table of content. Figures are given as the screenshots and some key information is not presented.

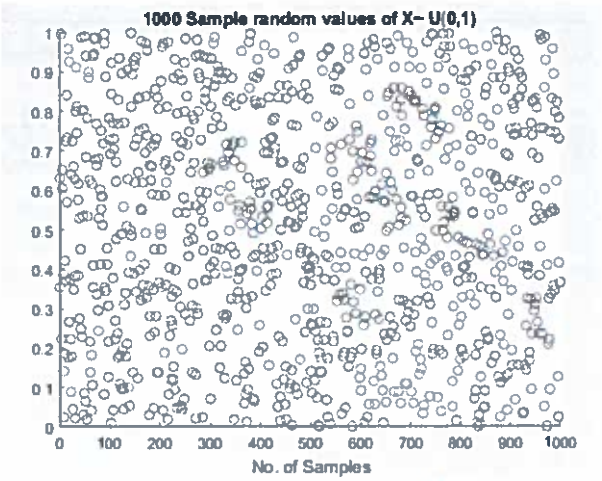
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CID: 00984531

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1. Random Signals and Stochastic Processes

Q 1



Screenshoot is not the best way figure illustration

Figure 1.1(a). Distribution of random values X~U (0, 1)

Q1.

The theoretical mean is calculated to be =0.5.

$$\int_0^1 xf(x)\,dx = \int_0^1 x\,dx = 0.5$$

The sample mean calculated by the mean function in MATLAB is: 0.51. This value differs from the expected value by

$$\left(\frac{0.51-0.5}{0.5}\right) * 100 = 2\%.$$

Star does not denote multiplication

However for a small sample size of only 1000 samples, this is acceptable.

Q2

The theoretical standard deviation for an ideal distribution is:

$$\frac{1}{\sqrt{12}}$$

Roughly=0.2887.

Correct values for theoretical/sample mean and std

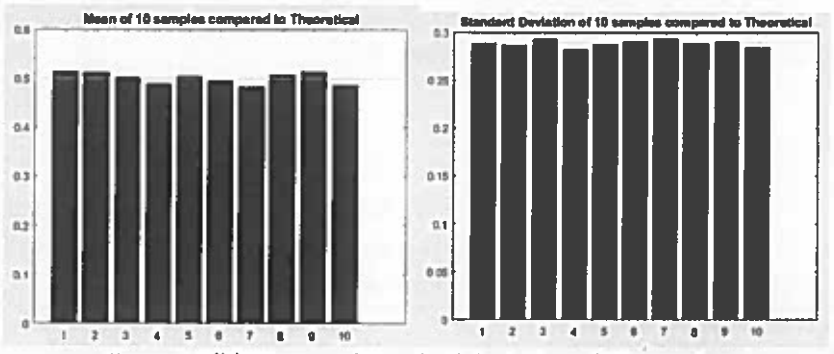
The standard deviation from matlab function std is 0.285.
Value differs by

$$\left(\frac{0.2887 - 0.285}{0.285}\right) * 100 = 1.3\%$$

This small difference is due to a small sample size of 1000 samples.

No Explanation of Sample Mean and sample std as Estimator

Q 3



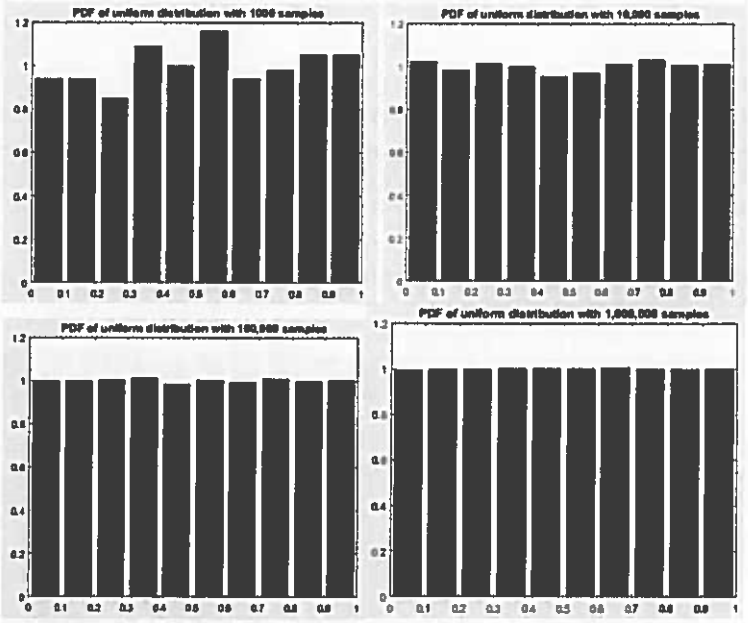
No explanation of bias

Legend is missing, figure are not vertically aligned. Also bar chart is not the best choice

Figure 1.1(b) Mean and standard deviation of 10 samples

From the bar charts above, the samples have an even distribution of mean and standard deviation values clustering around the theoretical value. The average of the means and standard deviations of the 10 sets of data are 0.4997 and 0.2883 respectively. There is evident improvement on the mean and standard deviation values as the bias is minimal. As we increase sample size we will get better values of mean and standard deviation with smaller bias.

Q 4



Theoretical pdf is missing

analysis of pdf with different number of bins is missing

Figure 1.1(c) PDF of uniform distribution of increasing samples.

The 1000 sample size and 10 histogram bins plot is a poor display of a uniform distribution. As we increase the sample size from 1000 up to 1,000,000, the probability density function gradually flattens out and we can see a good uniform distribution starting from 100,000 samples. Thus as sample size is

increased, the estimates will converge to their theoretical values. As more histogram bins are used, it is expected that there will be more outliers in our PDF plot with less samples but there will not be significant effect on PDF plots with high number of samples.

subsection labelling is inconsistent

Q5

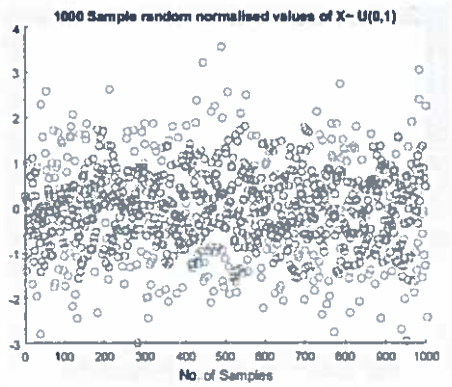


Figure 1.1(d) 1000 normalized values of (0, 1)

Q1.

The mean calculated by Matlab was 0.0560. For a small sample size like 1000, this is considered close to the theoretical mean of 0.

Don't place text beside figure

Not enough of explanations

Q2.

The standard deviation calculated by Matlab was 0.9887. For a small sample size like 1000, this is considered close to the theoretical standard deviation of 1 for a Gaussian random variable.

Q3.

Bias on Mean and Standard Deviation I collected data on the mean and standard deviation of 10 sets of 1000 samples using the randn function in MATLAB Using the data collected, the bar charts below were plotted to see how the values compared to the expected values of mean and standard deviation.

Don't use personal pronouns

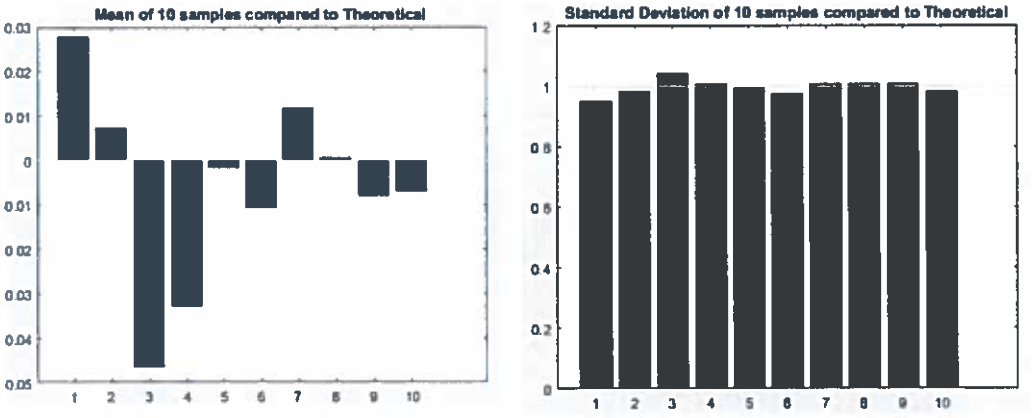


Figure 1.1(e) mean and standard deviation of 10 sample sets

The average mean and average standard deviation from the data collected is -0.0028 and 0.9889 respectively. The bias for both these values are small and will be smaller if we use a larger sample size.

Legend is missing

Q4.

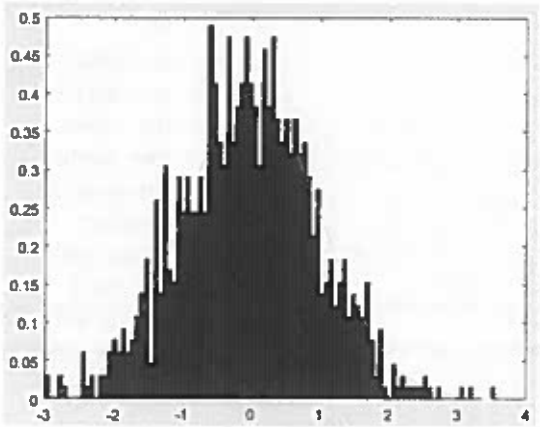


Figure 1.1(f) 1000 sample size PDF

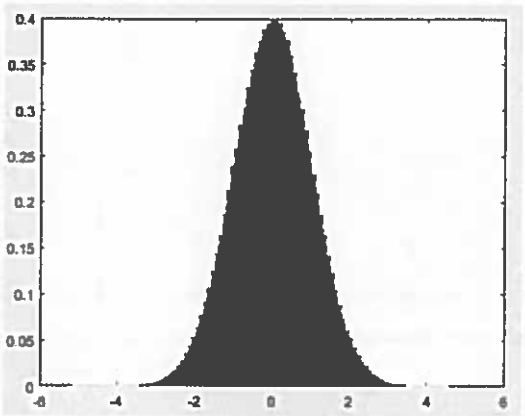


Figure 1.1(g) 1,000,000 sample size PDF

The plot from 1000 samples is not a good representation of a Gaussian random variable with mean 0 and standard deviation 1 as there is a lot of variation from the green theoretical line. 1000000 samples is used instead of 1000, and the results was a huge improvement. PDF plot for 1000000 samples is smoother and follows closely to the Gaussian curve. Therefore, as sample size is increased, the estimates will converge to their theoretical values. I have used more histogram bins in the above plots as the range of the data is more than the previous data generated with the rand function. This is to give a better picture of the full Gaussian plot.

1.2 Stochastic Processes

Bad format of figures

Q1.

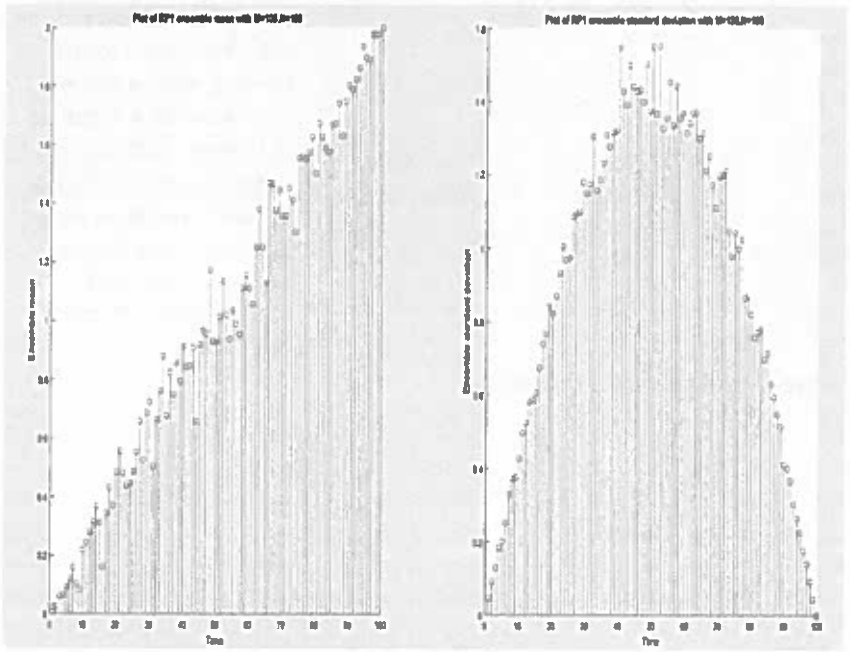
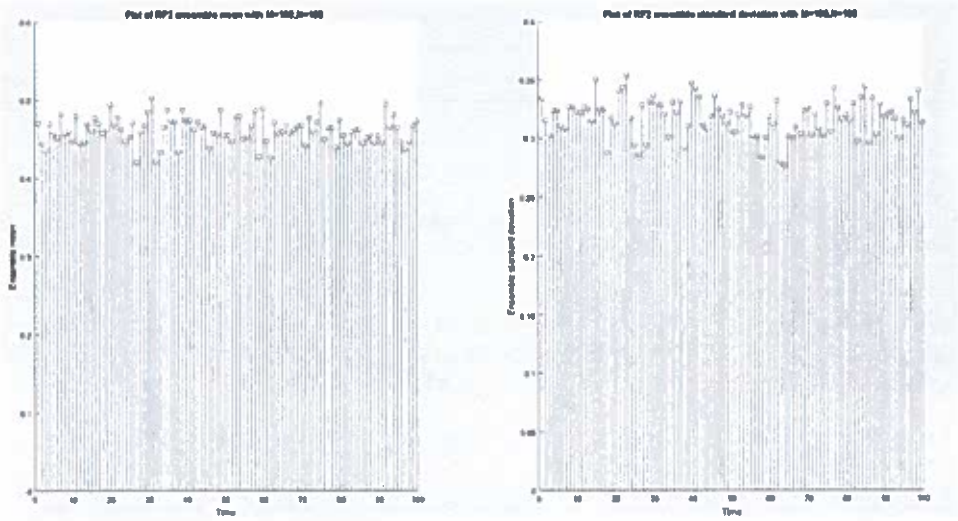


Figure 1.2(a) RP1 ensemble mean and standard deviation

RP1.

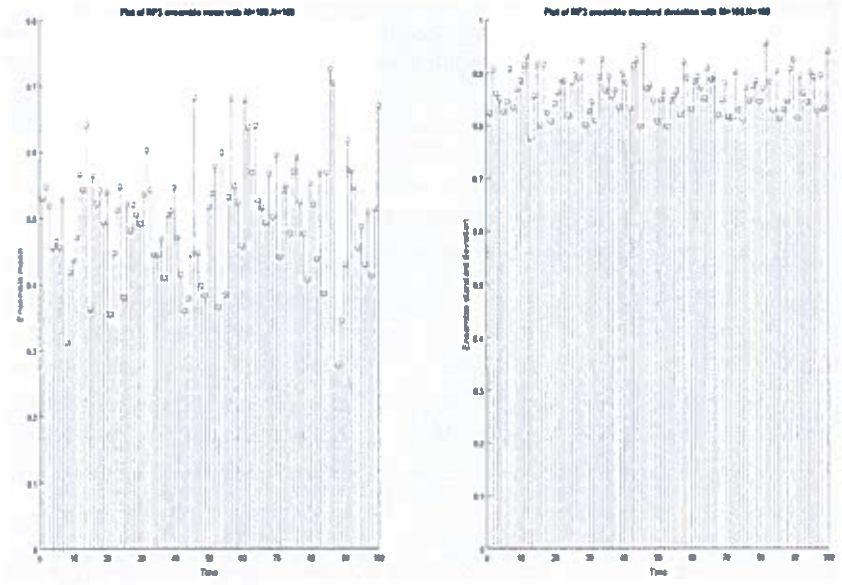
The mean and standard deviation are both not stationary. The mean increases linearly overtime with gradient 0.02 but with more fluctuations around midrange of the plot. This property coincides with the standard deviation curve that shows higher values around the midrange. The standard deviation however, shows a sinusoidal path.

No definition of stationarity is given



RP2.
Stochastic process RP2 has values that cluster closely to a constant mean and standard deviation so it is stationary.

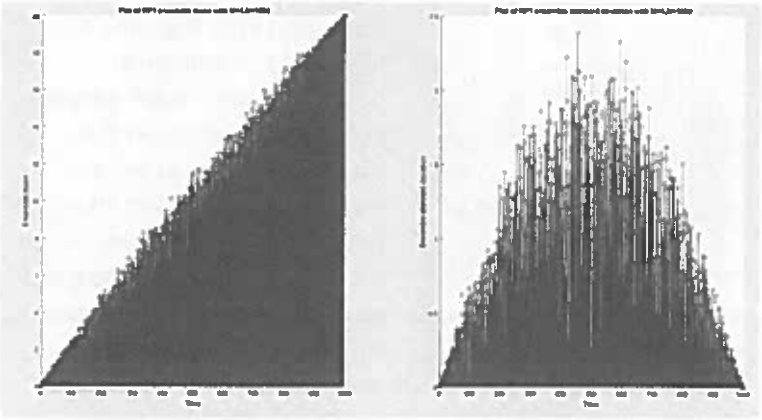
Figure 1.2(b) RP2 ensemble mean and standard deviation



RP3.
The standard deviation is stationary and slightly varies around its constant standard deviation whilst the mean varies with much more fluctuations around 0.5. The reason for this is because the variance is multiplied by a factor of 3, thus the data will vary more but still clustering around a constant value. Therefore, we assume that the mean and standard deviation is stationary.

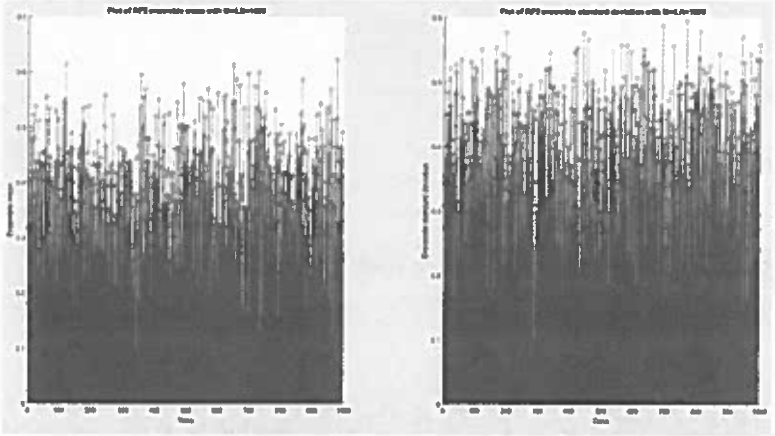
Figure 1.2(c) RP3 ensemble mean and standard deviation

Q2.RP1.



RP1 is not ergodic because the ensemble average and standard deviation changes overtime.
And once again the fluctuations in the midrange of the ensemble mean plot coincides with the higher values of standard deviation around the midrange.

Figure 1.2(d) RP1 ensemble mean and standard deviation for M=4 and N=1000.



RP2.

RP2 is ergodic as the ensemble mean and standard deviation are constant but fluctuations are severe due to a high standard deviation but the data still cluster around a constant value overtime.

Figure 1.2(e) RP2 ensemble mean and standard deviation for M=4 and N=1000.

Definition of ergodicity is not given. Incorrect identificaiton of ergodicity for random processes

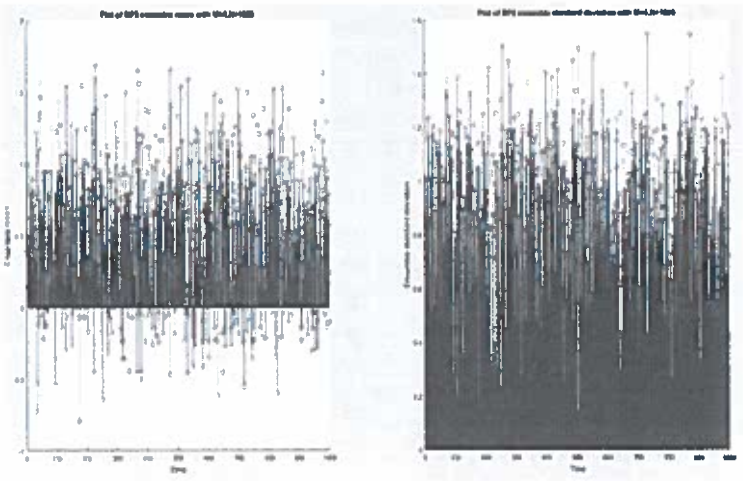


Figure 1.2(f) RP3 ensemble mean and standard deviation for M=4 and N=1000.

RP3.
As we can see, fluctuations in our plot is very high due to a high value of standard deviation. The reason why we get negative values in the mean is because as we are using less realizations, the data is less averaged out, thus with a huge standard deviation, we will have more severe outliers. But overall RP3 is still ergodic as the data cluster around a constant value overtime.

Q3

No derivation is provided

Theoretical mean of RP1

$$E\{rp1\} = E\left\{(u(n) - 0.5) \times 5sin\left(\frac{n\pi}{N}\right) + 0.02n\right\} = 0.02n$$

Standard Deviation of RP1

$$Var\{rp1\} = E\left\{\left[(u(n) - 0.5) \times 5sin\left(\frac{n\pi}{N}\right)\right]^2\right\} - E\{(0.02n)^2\} = \frac{25}{12}\left[sin\left(\frac{n\pi}{N}\right)\right]^2$$

$$SD = \sqrt{Var\{rp1\}} = \frac{5}{\sqrt{12}} sin\frac{n\pi}{N}$$

The standard deviation is a sine wave with amplitude $\frac{5}{\sqrt{12}}$. This corresponds to our plot in Figure 1.2(d) that has an amplitude of 1.45.

Theoretical mean of RP2

$$E\{rp2\} = E\{(u(n) - 0.5) \times U(M) + U(A)\} = 0.5$$

Standard Deviation of RP2

$$Var\{rp2\} = E\{[(u(n) - 0.5) \times U(M) + U(A)]^2\} - E\{0.5^2\} = \frac{1}{9}$$

Theoretical mean of RP3

$$E\{rp3\} = E\{3[U(n) - 0.5] + 0.5\} = 0.5$$

Standard Deviation of RP3

$$Var\{rp3\} = E\{[3[U(n) - 0.5] + 0.5]^2\} - E\{0.5^2\} = \frac{3}{4}$$

$$SD = \sqrt{Var\{rp3\}} = \sqrt{\frac{3}{4}} = 0.866$$

Analytical results are not compared with the sample mean and std

1.3 Estimation of Probability Distributions

Q1.

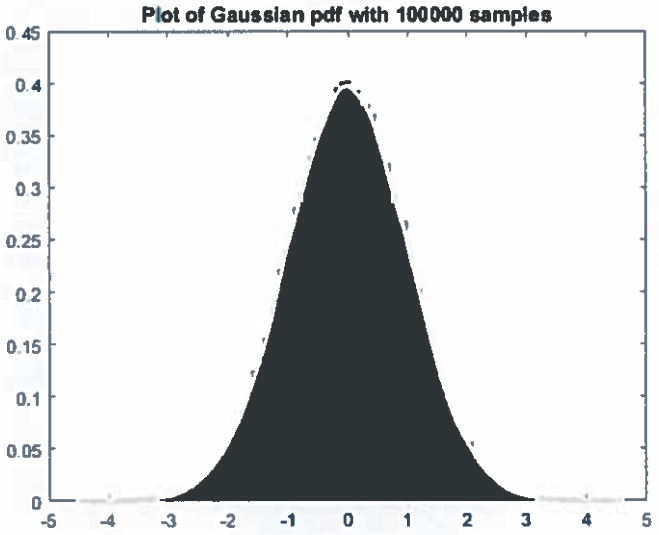
Function pdf was coded as follows:

code of pdf is provided

```
function [y] = pdf(samples)
 hold on
bin = 100;
[y,x]=hist(samples,bin);
bar(x,y/trapz(x,y));
 [ref, z] = ksdensity(samples);
plot(z,ref,'g','LineWidth',3) ;
hold off
```

More analysis is needed

Using a Gaussian variable with 100,000 samples, the following graph was plotted:



Legend is missing

Figure 1.3 (a) Gaussian pdf with 100,000 samples.

RP2 is not stationary Therefore, it shouldn't have been included

Q2.

Legend is missing

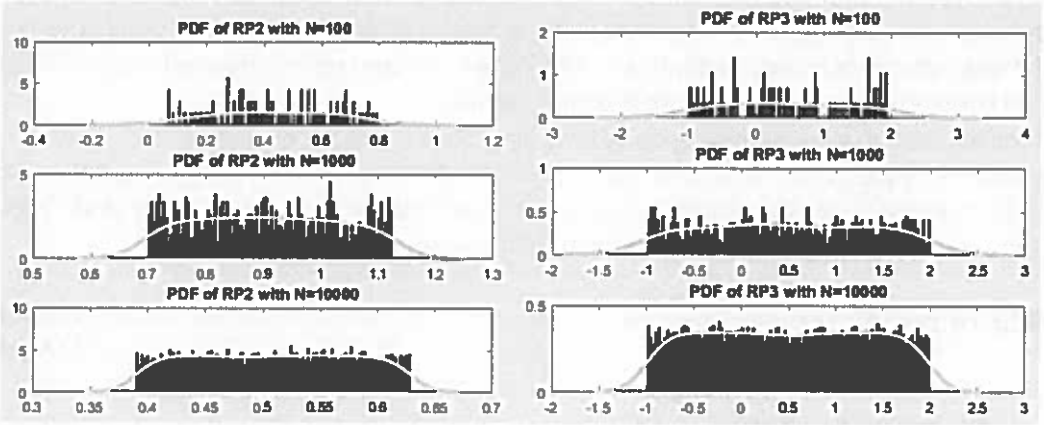


Figure 1.3 (c) Pdf of RP2 & 3 with increasing samples.

The higher the number of samples, the closer the approximation is to the theoretical PDF.

Q3.

Due to changing characteristics such as mean for non-stationary processes, my function will not be suitable. When the mean changes from 0 to 1 when N=500, the PDF will shift to the right. To accommodate for this changing mean with N, I will code for a changing PDF function that has an additional parameter N.

No sensible solutions to compute pdf of non-stationary processes

No explanation of the difficulty computing the mean of the non-stationary process

2. Linear stochastic modelling

2.1 ACF of uncorrelated sequences

Q1. & Q2.

Correct ACF estimate of WGN, however, the figure is badly formatted

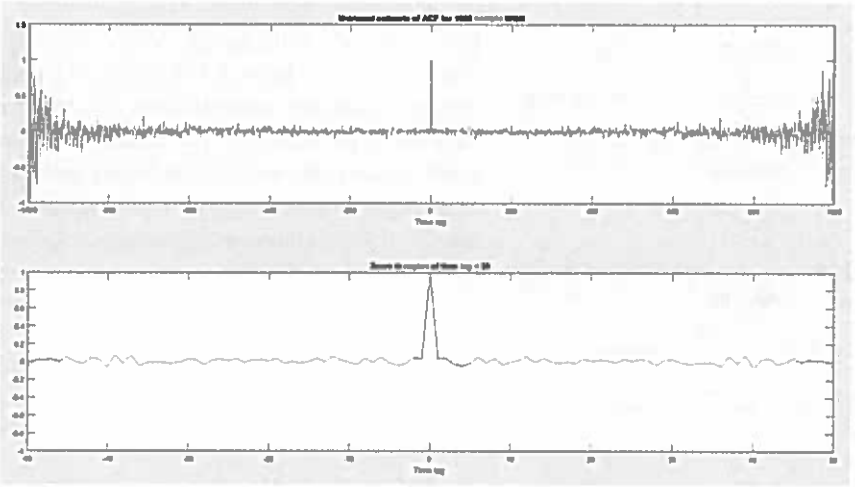


Figure 2.1 (a) Plot of unbiased estimate of ACF for 1000 sample WGN and zoomed in view of the plot

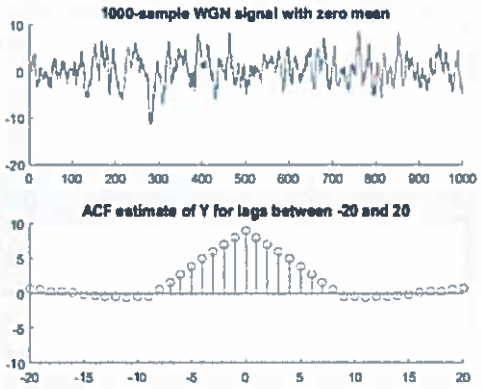
Some explanation for different sizes of lag is given

This function has the form of a discrete Dirac function at time lag =0, and since it is normalised, we obtain a correlation of 1. We can conclude that the ACF is an even function as it is symmetrical on both sides of 0. This is due to the AC matrix being symmetric and Toeplitz.Due to WGN being independent and is identically distributed, it is perfectly correlated to itself (at zero lag) and completely uncorrelated with all other lag values.
From the zoomed in view of the region of time lag < 50, as time lag gets closer to 0, the correlation increases, peaking at time lag = 0. We can see there are increasing oscillations at time lags of above |700|. Conclusively, estimates of ACF higher than 700 or lower than -700 time lags are not statistically reliable as the function is not stationary.

2.2 ACF of correlated sequences

Correct plot of ACF of MA process

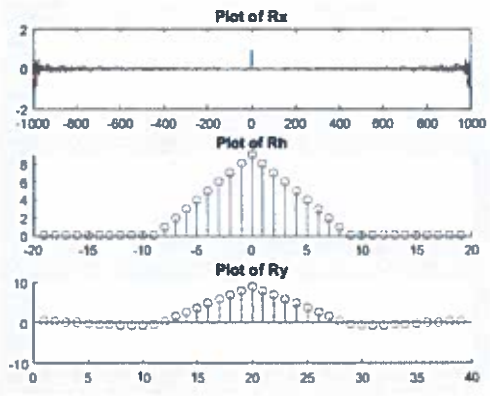
Q1.



Since the filter which we use has 9 coefficients which are all ones, we are essentially only using 9 samples for our calculation of ACF and thus, we only observe 9 decreasing peaks from time lag = 0 to 8. The ACF for higher lags is zero. Sample mean can only be calculated after time lag of 9 or less than -9. This is because the processes only becomes stationary when ACF is 0.

Figure 2.2 (a)

Q2.



For Rx, the ACF function of a WGN is an impulse response at time lag = 0. For Rh, its ACF is similar to the previous question with MA filter (9). After convoluting Rx and Rh, the ACF plot still has a peak at time lag=0 and starts to decrease as time lag increases but at a slower rate than ACF of Rh, as it only reaches zero after time lag=29. If Xn is an uncorrelated process, Ry represents the ACF of the MA filter.

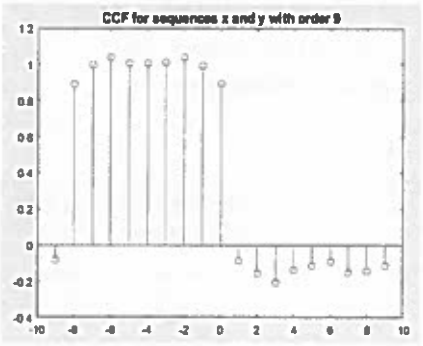
Figure 2.2 (b)

No mathematical proof

2.3 Cross-correlation function

Q1.

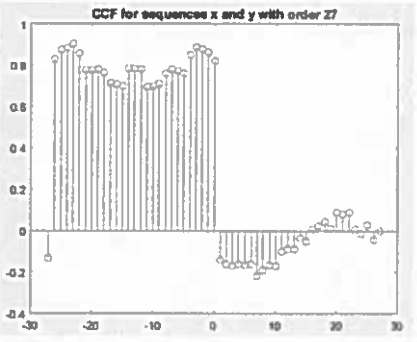
Correct plot of CCF of MA process



The 9 large pulses with magnitude 1 from time lag=-8 to 0 corresponds to the filter order of 9. As Xt is an uncorrelated stochastic process and as the ACF of x is the Dirac delta function (a spike of magnitude 1 at zero lag), convoluting the filter impulse response with this only outputs the impulse response 9 times.

Figure 2.3 (a) Caption is missing

Bad layout



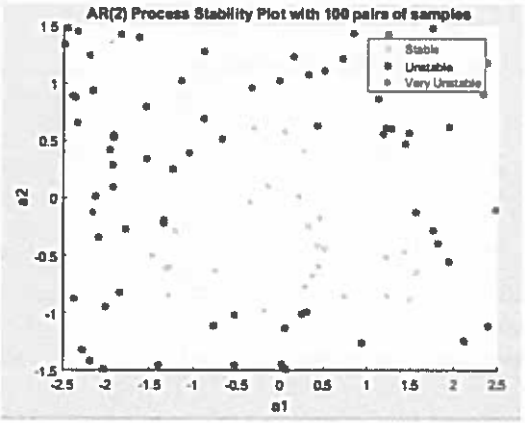
We can identify the order of the filter using the number of large pulses. Using order 27 for the filter, as expected 27 large pulses were seen at the cross correlation.

Figure 2.3 (b)

No explanation of the shape of Cross correlation for white input

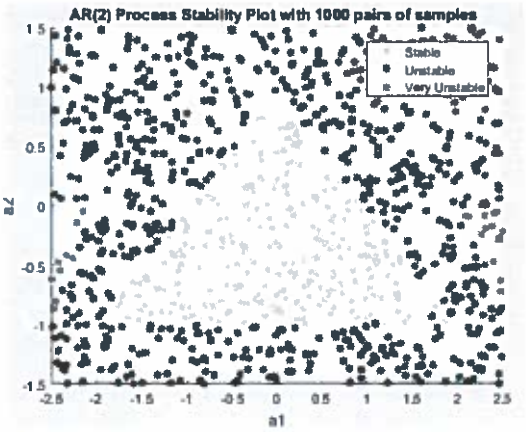
2.4 Autoregressive modelling

Q1



Stable processes are situated in the middle of the plot but doesn't give a distinct shape. A plot using 1000 samples was generated to get a better shape of the region of convergence.

Figure 2.4 (a)



The plot exhibits symmetry around a1=0, this suggest that the function is even. We can also see a more distinct triangular region of convergence. The constraints for a1 and a2 for stability are:

1. $a_1 + a_2 < 1$;
2. $a_2 - a_1 < 1$;
3. $|a_2| < 1$;

No derivations are provided

Figure 2.4 (b)

Q2.

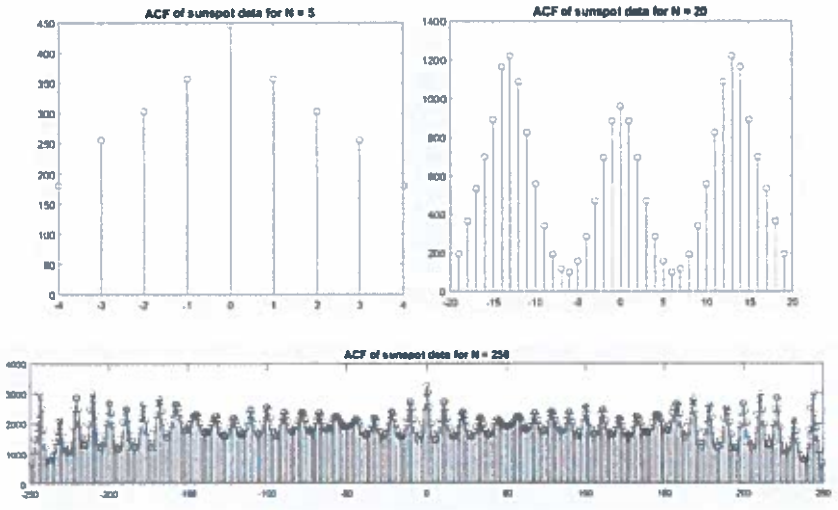
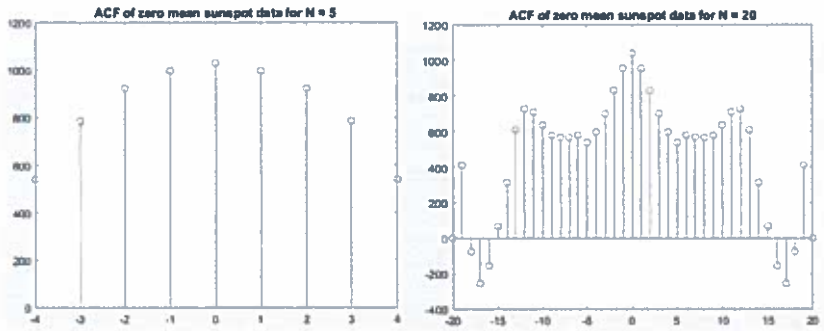
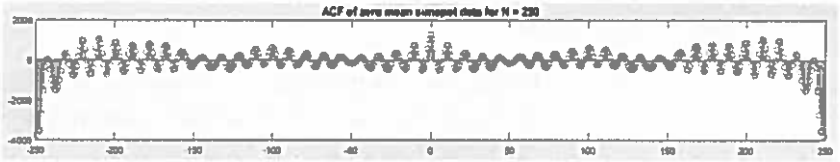


Figure 2.4 (c)



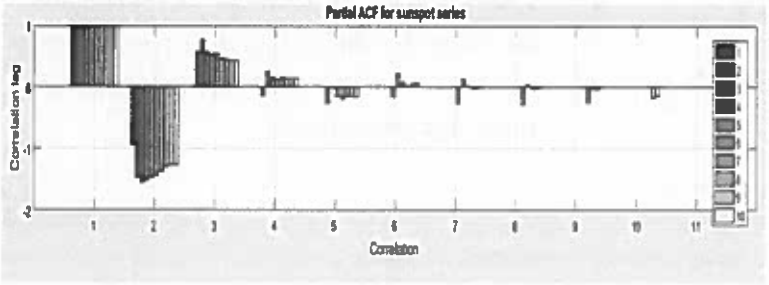


Some comment on the differences is presented

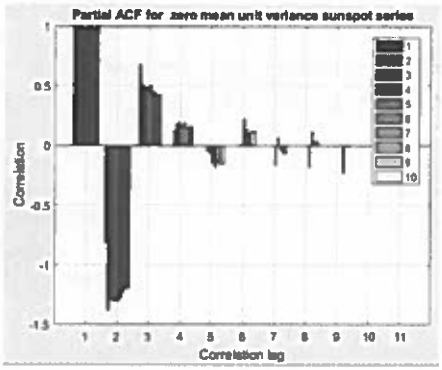
Figure 2.4 (d)

Figure 2.4 (c) is for the real world sunspot series, and Figure 2.4 (d) is with zero mean.
For Figure 2.4 (c), the series is all positively correlated suggesting an increasing trend, after zeroing the mean (Figure 2.4 (d)), the ACF plot oscillates around zero.
For N =5,
We see a decrease in ACF away from the origin, and the ACF peak value is much less than those with larger N.
For N = 20,
We see a sinusoidal pattern emerging with peaks happening at 13 time lags apart.
For N = 250,
We see a sinusoidal pattern emerging with peaks happening at 13 time lags apart.
The ACF increase from a peak of 500 at n =5 to 1200 and to 3000 for n =250.
The mean acted as an off set to our ACF plot, making it all positive. We can extract more information when it is removed such as when the ACF cross 0 when which parts are negatively correlated.

Q3.



Bad format of the legend

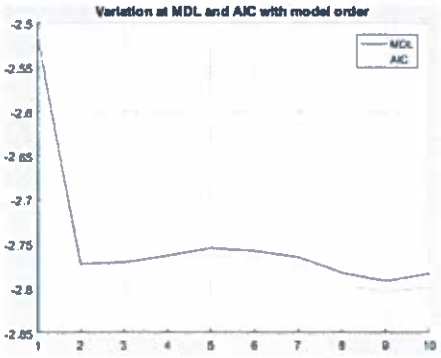


The best order of the sunspot series is a 2nd order AR process. Notice that the 1st 2 coefficients are the ones that affect the ACF the most.

Correct Identification of model order

Captions are missing

Correct plot of MDL and AIC



Q4.

Figure 2.4 (e) is the MDL and AIC for model orders up to 10.The plot shows a significant decrease in error after order 2. From order 3 to 5 we see a slight increase in error and from 6 to 9 we see a slight decrease in error. The ideal model order would be 2. This is because to even have a slight decrease in error after model order 2, the order needs to be at least above 8. Thus there is no point going above order 2 as this will only increase complexity and costs with no significant benefits.

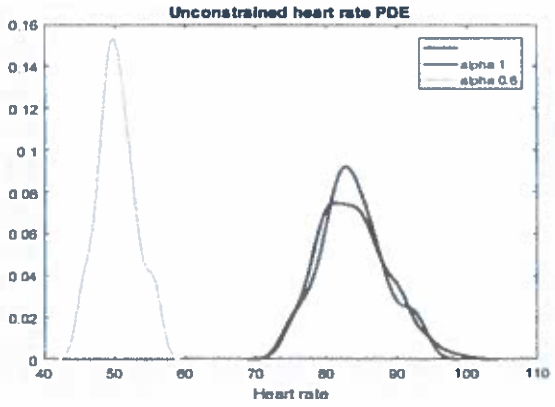
Figure 2.4 (e) Variation of MDL & ACL with model order

Part of the assignment is missing

2.5 Real World Signals: ECG from iAmp Experiment

Q (a) & (b)

plot of the probability density estimates is given



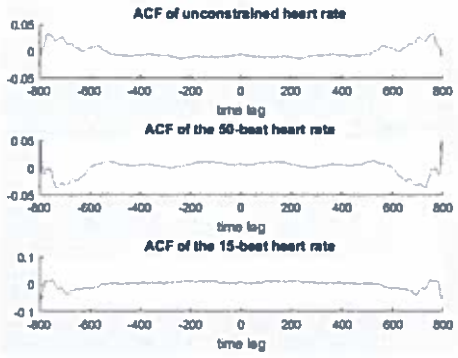
The average heart rate PDE is plotted in blue and it takes a normal distribution form with mean=82. When alpha = 1, the PDE of the averaged heart rate has the same form but the range of values it takes decreased, area under the curve stays the same but as values gets nearer to mean, its probability increases. For alpha=0.6, the curve is shifted and the new mean is 50. Its range of values taken also decreased.

Figure 2.5 (a) PDE of unconstrained heart rate.

Explanation is not complete

AR modelling of heart rate

Q(c)

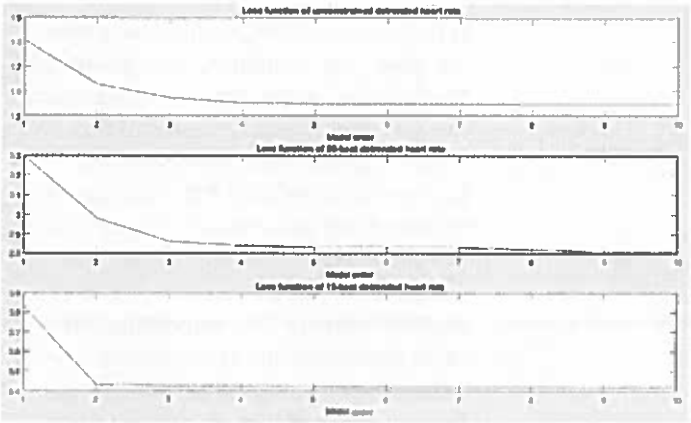


The shape of the three trials resembles the zero mean ACF of an AR (2) process that we covered in section 2.4.

Correct plot of the ACF of the three trials However, the format is bad

Figure 2.5 (b) ACF of heart rates.

Q (d)



AR model order 3 or 4 is suitable for my unconstrained heart rate. While a AR model order 3 would suit the 50beat heart rate, and a 2 order AR model would suit the 15beat heart rate.

Figure 2.5 (c) Loss function of heart rates

3. Spectral estimation and modelling

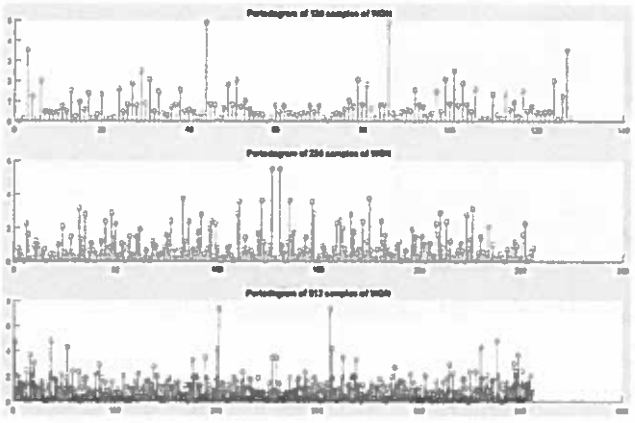
Matlab code for pgm:

No definition of PGM acronym

```
function [y] = pgm(x)

N = length (x);
for a =0:(N-1);
    sum =0;
    for n=0:(N-1);
       sumtemp = x(n+1,1)* exp ( -1 * 1j * 2 * pi * a * n/N);
       sum = sum + sumtemp;
    end
    y(1,a+1) = ((abs(sum))^2)/N;

end
end
```



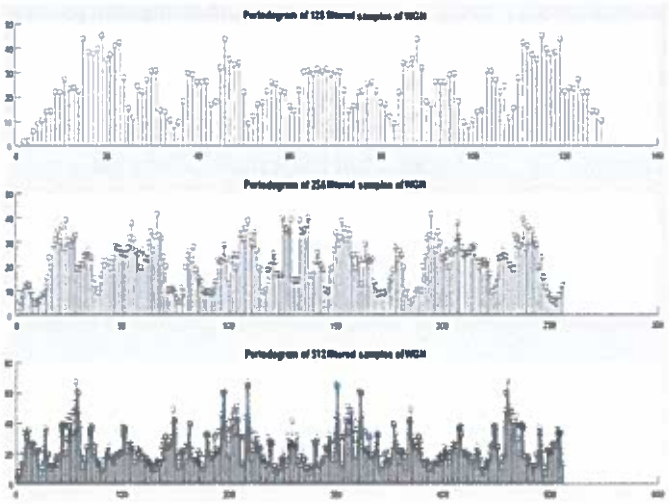
PSD estimate of WGN is symmetrical about N/2. Even though in theory, the PSD of a WGN should be constant but it is hardly the case here as the plotted PSD estimate fluctuates a lot. This is due to our sample's finite length.

Some explanation of periodogram result for white noise

Figure 3(a) Periodogram of WGN with different sample size.

3.1 Averaged periodogram estimates

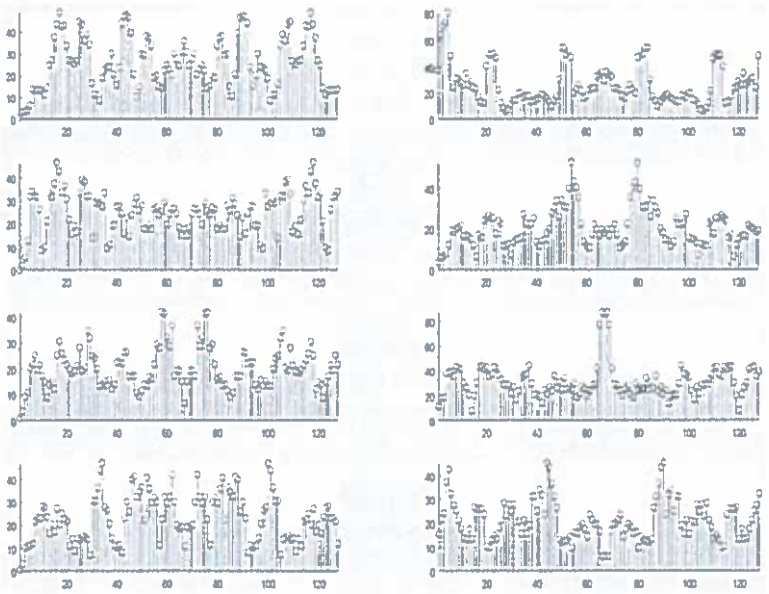
Bad format of the figure



Q1. After smoothing with a zero phase FIR filter, high frequency components of the function were removed, thus making the estimate smoother. Most of the extreme values that comes with the outliers in our original PSD estimate were removed and even though the plot does not show a constant unity for all frequencies, it is still smoother than the original estimate. The smoothing effect is most obvious on our lower sample size of 128. This can be explained as there is a finite number of frequency bins to be used for dispersing power.

Figure 3.1 (a) Frequency averaged periodogram for different sample sizes.

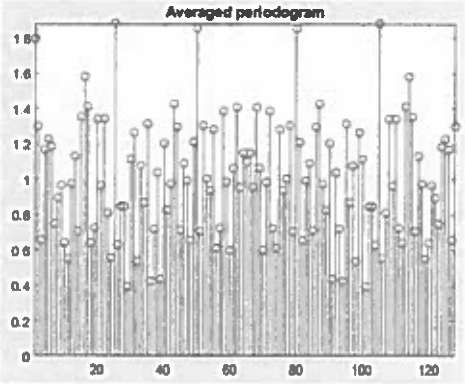
Q2.



As we learned from the previous question, a smaller sample size will allow the filter to work better. The samples vary randomly, they either take up values from 0 to 40 or 0 to 80. Averaging these 8 results might give a better estimate.

Figure 3.1 (b) Subdivided 128- filtered sample segments of a 1024- sample sequence of WGN.

No explanation of the similarities and variation of each periodogram



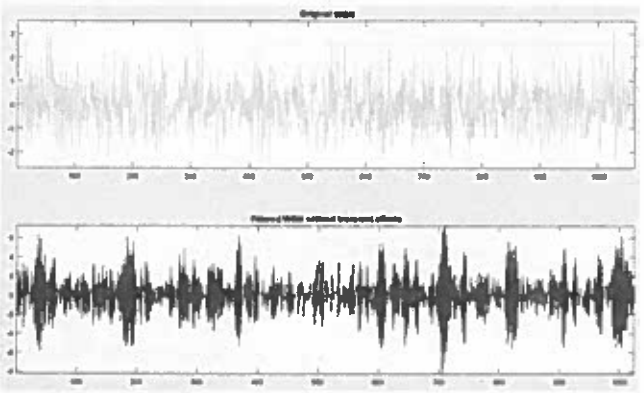
Q3.

The averaged PSD also shows less variance and closer resembles the theoretical PSD of WGN.

Stem plot is not the best choice

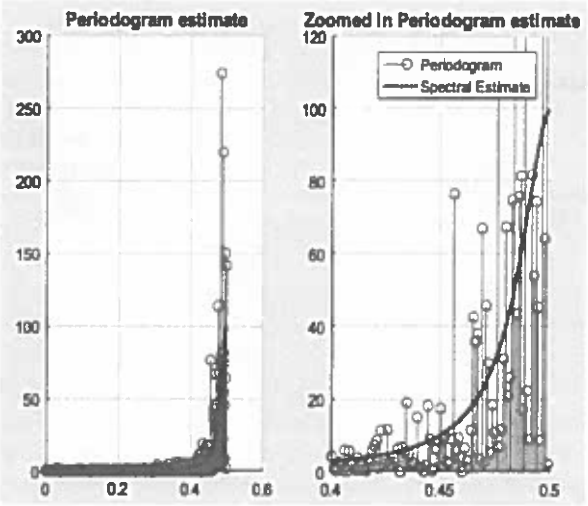
Figure 3.1 (c)

3.2 Spectrum of autoregressive processes



Some explanation of high pass nature is given

Figure 3.2 (a)



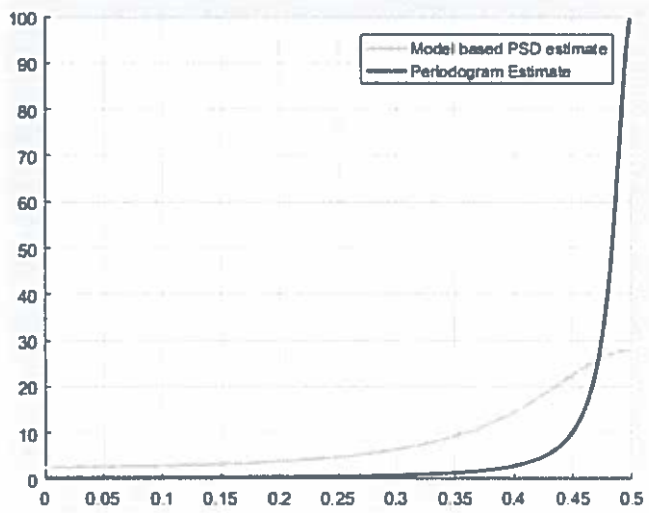
As we can see, the shape of the spectral estimate is exponential as it increases rapidly from 0.4 to 0.5, due to it being a high pass filter. The cut-off frequency is approximately 0.47 Hz. The difference in the red and blue plots are apparent in the region above 0.4 where there are high spectral components that appear above the blue line. The results are different due to the input signal is of finite width and since we are applying a rectangular window, it introduces high frequencies due to its discontinuities at its edge.

Figure 3.2(b)

Caption is missing

Variance is not explained

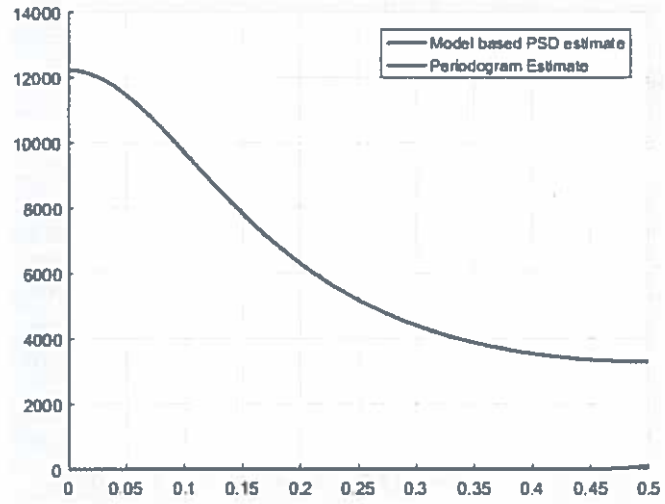
Q4.



The model based PSD estimate gave a similar shape although with less of an exponential rise.

Figure 3.2 (c)

Q5.



The periodogram has a similar shape as the model based PSD of model order 2. Over modelling will cause the PSD estimate to oscillate while under modelling will cause the PSD estimate to be inaccurate.

No reasonable explanation on this topic is given

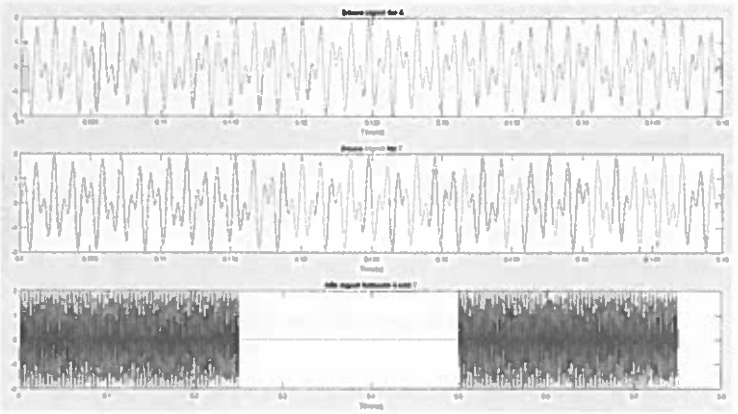
Figure 3.2 (d)

3.3 Spectrogram for time-frequency analysis: Dial Tone Pad

```
london_num = randi([0 9],1,11);
london_num(1,1) = 0;
london_num(1,2) = 2;
london_num(1,3) = 0;
fs = 32768;
pressT = 0:1:0.25*fs-1;
idleT = zeros(1,0.25*fs);
f1=[1336 1209 1336 1209 1477 1477 1209 1477 1477 1477 1209];
f2=[941 697 941 697 852 852 697 852 770 697 852];
y = 0;

for n = 0:10;
y1  = sin(2*pi*f1(n+1)*pressT/fs) + sin(2*pi*f2(n+1)*pressT/fs);
y = [y y1];
end
final = 0;
for n = 0: 10;
    k = london_num(1,n+1);
    z = y(k*8192+1:(k+1)*8192);
    final = [final z idleT];
end
```

A higher sampling rate would be limited by the rate of pressing the keys. The sampling rate used is appropriate.



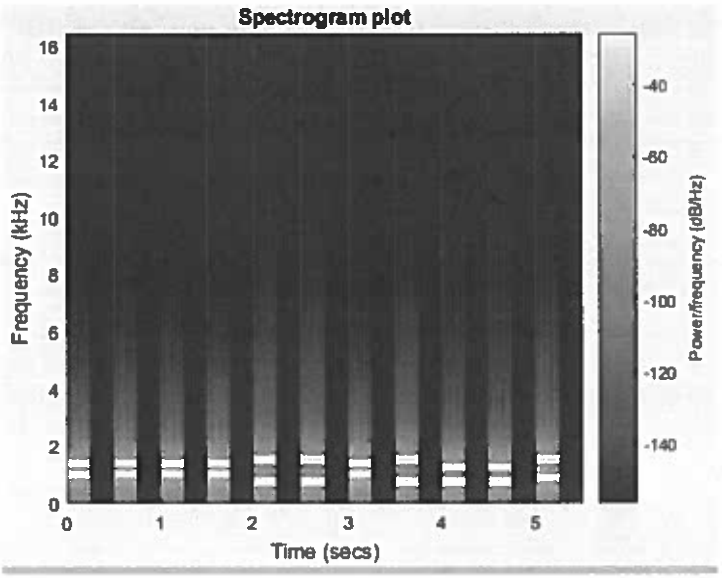
Generated landline: 0 2 0 2 9 9 2 9 6 3 7.

Using this number, the following spectrogram plot was generated:

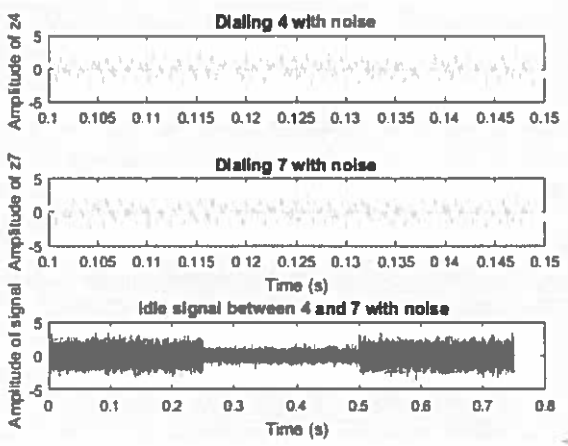
No time domain plot of the phone number is given

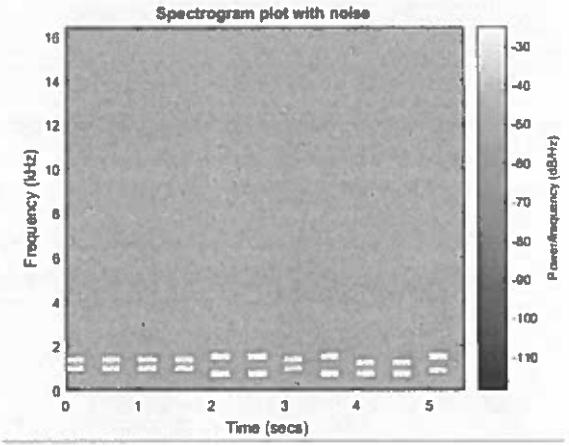
Axis scale could've been better

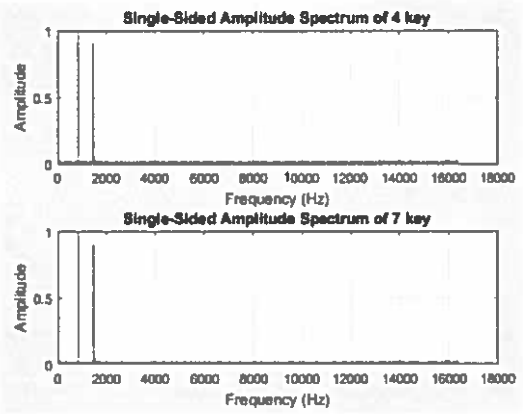
No explanation of key classification using spectrogram and table



Low noise

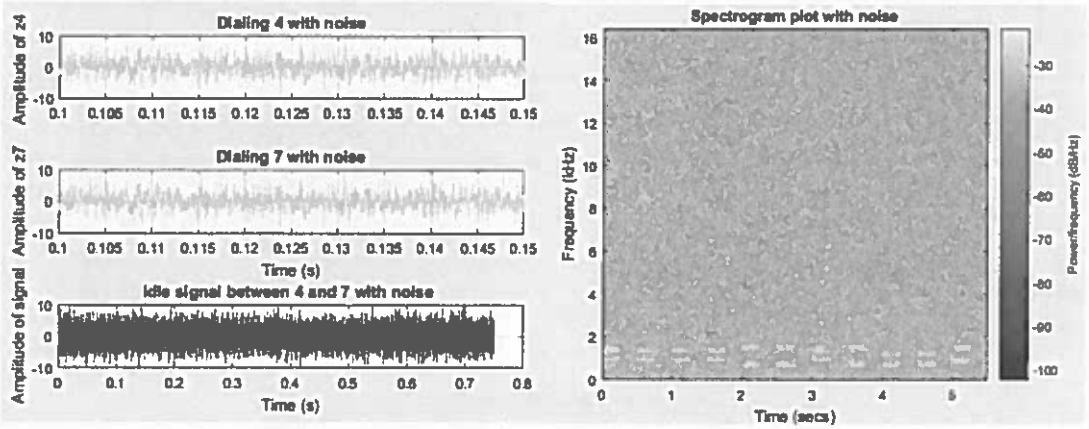


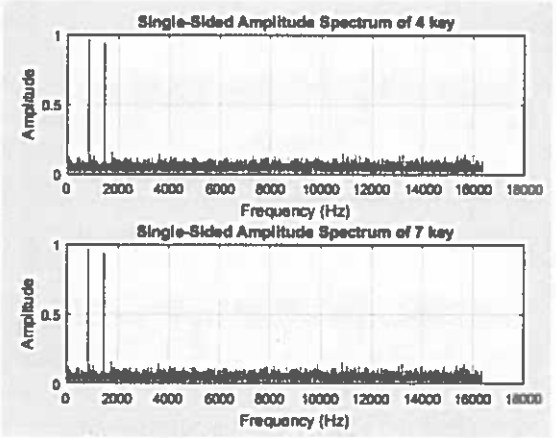




For low noise variance of 0.5, we can still discern the original waveform. A low noise variance of 0.5 will not affect our ability to distinguish keys as we can still find the 2 frequencies where spikes occur in the fft plot. Using the SNR formula, we can see that the SNR = 1/0.5 = 2 which is high, and we are able to identify keys with ease.

Medium noise

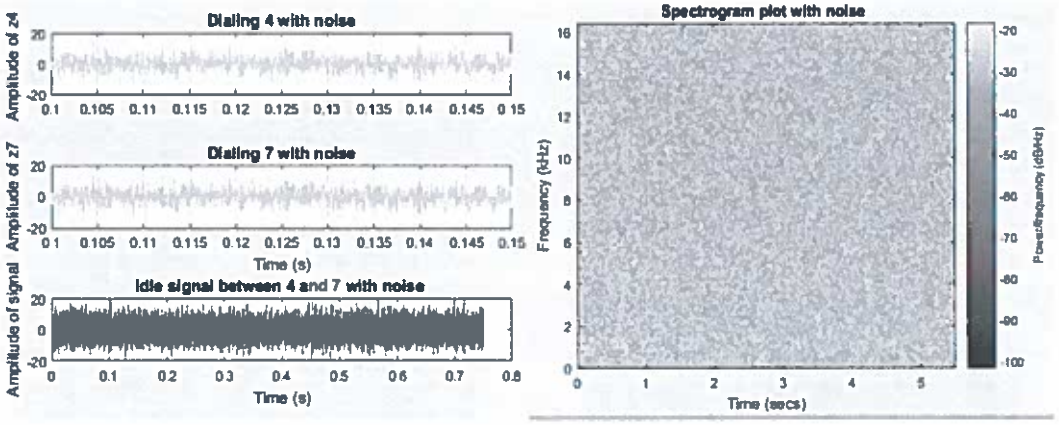




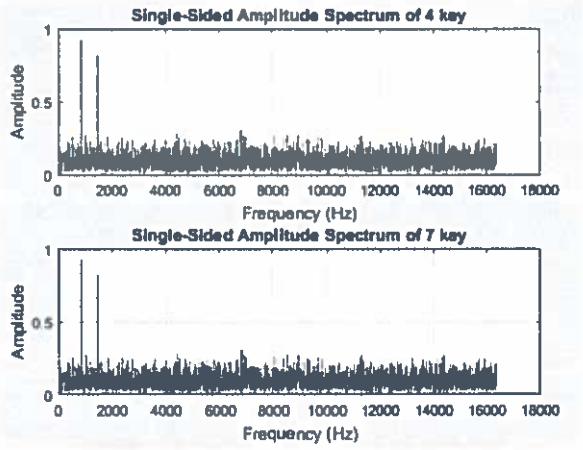
Caption is missing

For noise variance of 2.5, the signals look visibly distorted but it is still possible to distinguish the keys in the spectrogram plot and also make out the 2 spikes in the fft plot for 4 & 7.
The SNR = 1/5 = 0.2 which is quite low, but still sufficient for us to identify keys.

High noise



Caption is missing



For noise variance of 5, we can see that the signals are indistinguishable,
And it is impossible to identify which keys are pressed. The spectrogram is completely immersed in noise.
Using the SNR formula, we can see that the SNR =1/50 = 0.02 which is very low, and as expected we are unable to identify key presses.

3.4 Real World Signals

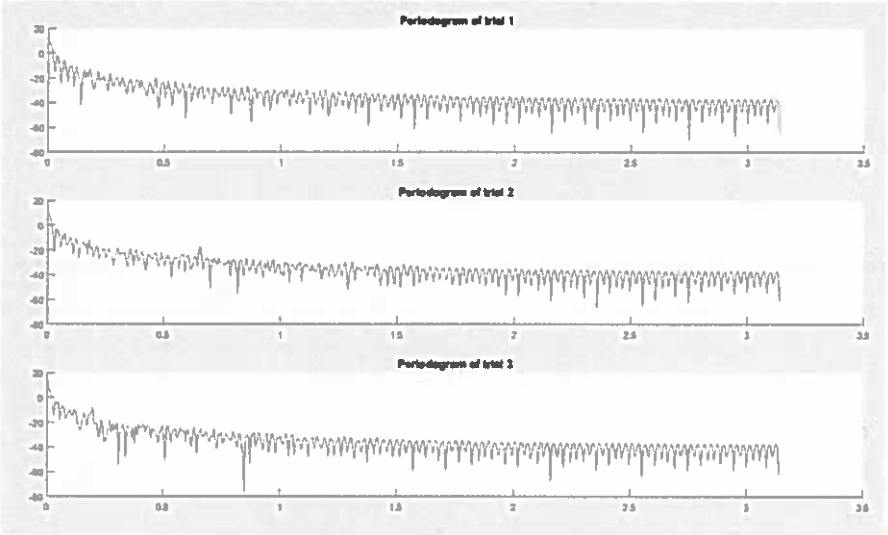


Figure 3.4 (a) Standard periodogram of all 3 trials

This part of the assignment is basically missing

A lot of blank space

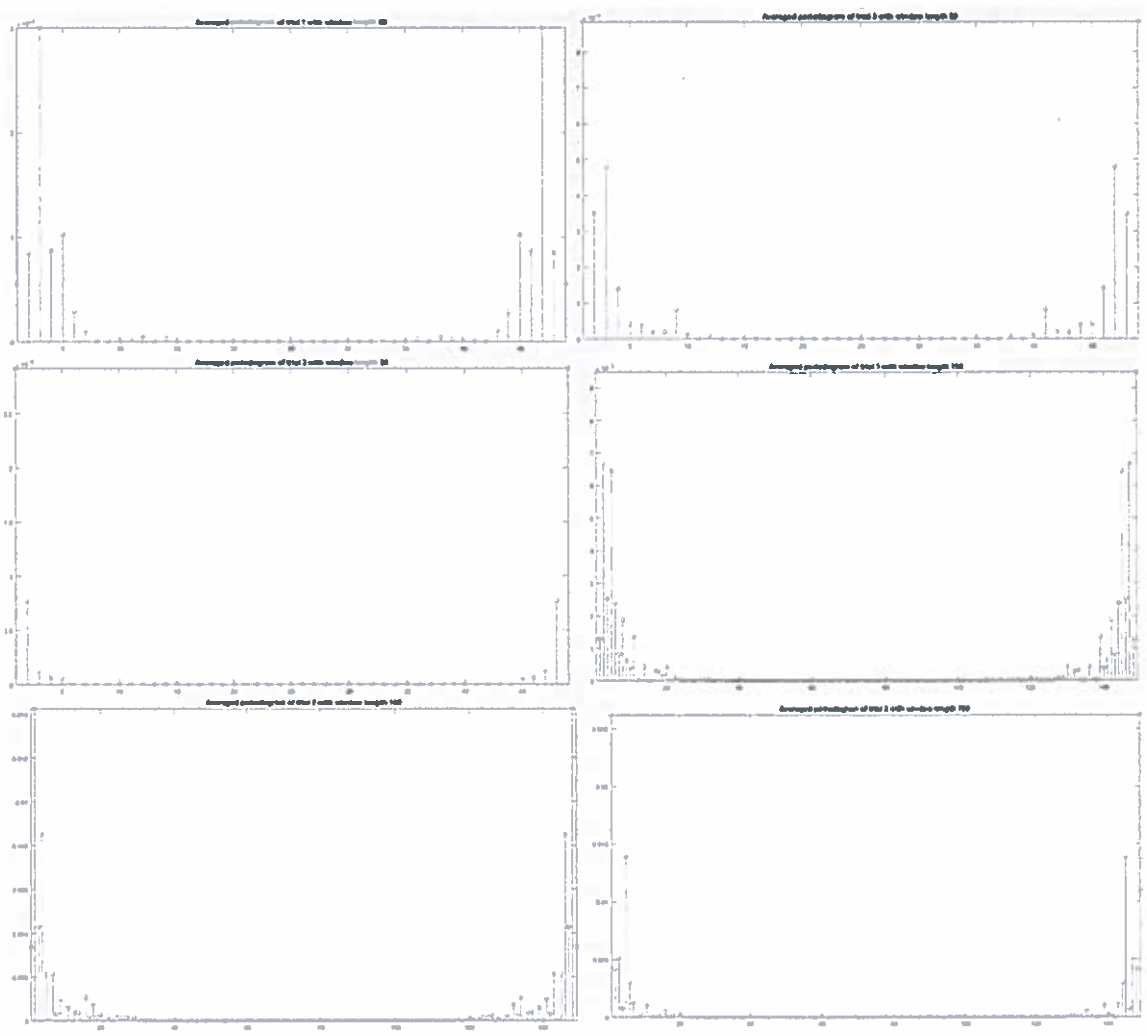


Figure 3.4 (b) Averaged periodogram with different window lengths.

b) Large spectral components corresponds to high heart rate whereas low spectral components corresponds to low heart rate. Trial 2 and trial 3 for constraint breathing shows that the person is taking faster shallower breaths. The pattern of breathing, taking long deep breaths, in trial 3 causes the test subject's heart rate to increase significantly.

4. Optimal filtering fixed and adaptive

4.1 Wiener filter

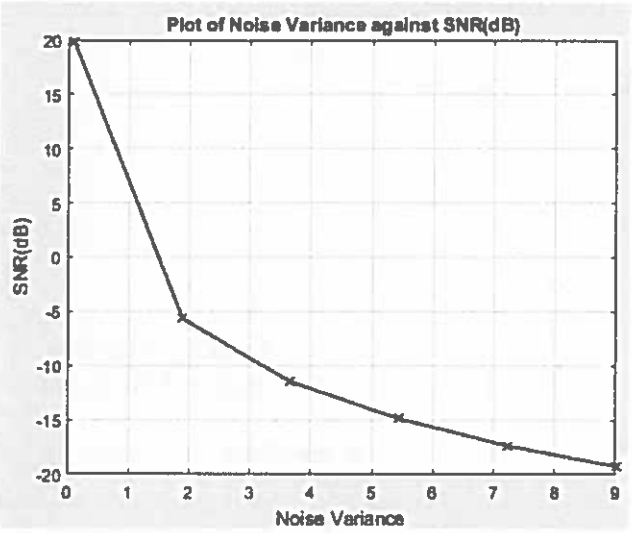
Theoretical SNR ratio is equal to $1/0.1^2$ =20 dB. My SNR for z[n] is 19.600 dB as calculated in matlab.

Rxx									Pzx	Opt. Soln.	Unknown system
1.0865	-0.0707	-0.0216	-0.0090	0.0095	-0.0066	-0.0284	0.0084	0.0002	0.0625	1.256	1
-0.0707	1.0865	-0.0707	-0.0216	-0.0090	0.0095	-0.0066	-0.0284	0.0084	0.1032	1.812	2
-0.0216	-0.0707	1.0865	-0.0707	-0.0216	-0.0090	0.0095	-0.0066	-0.0284	0.4194	3.408	3
-0.0090	-0.0216	-0.0707	1.0865	-0.0707	-0.0216	-0.0090	0.0095	-0.0066	0.3982	1.129	2
0.0095	-0.0090	-0.0216	-0.0707	1.0865	-0.0707	-0.0216	-0.0090	0.0095	0.2737	0.523	1

bad format of the table

Obtained Solution is not optimal.

Q2.



Correct SNR calculation

SNR decreases significantly as noise variance increases.

Q3.

To find Pzx and Rxx, $2Nw^2$ operations are used
To find the inverse of Rxx, Nw^3 operations are used.
Multiplication uses another Nw operations.
Hence to find optimal wiener solution, **$Nw3 + 2Nw^2 + Nw$** operations are required.

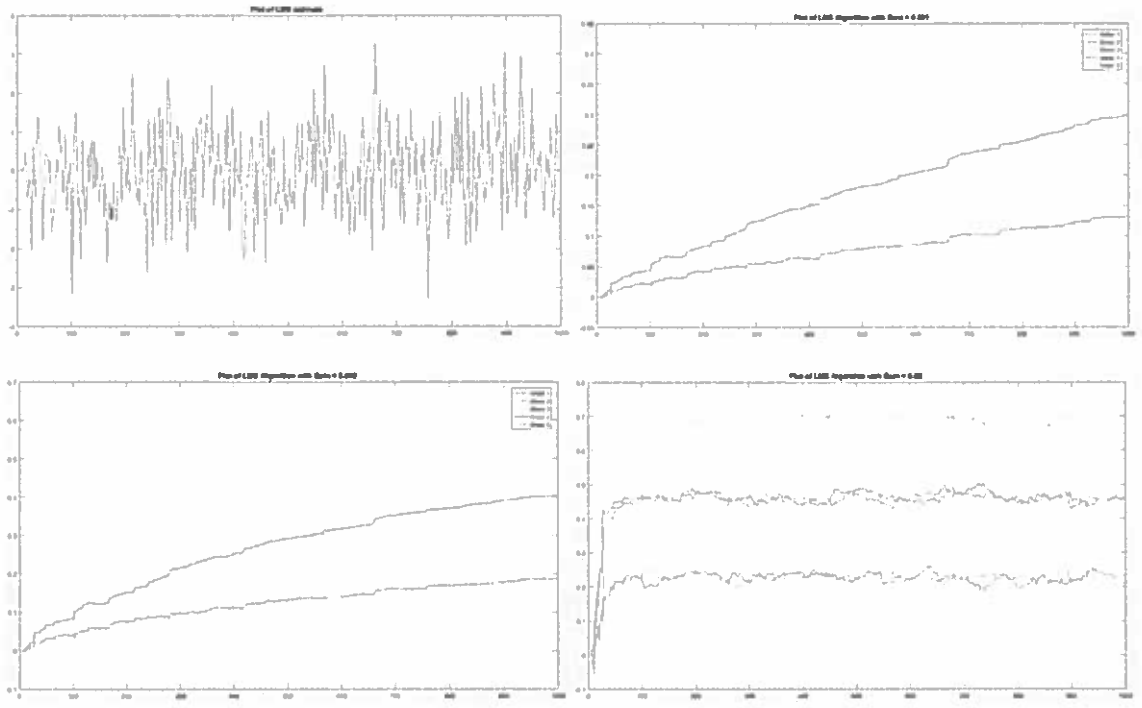
Some complexity analysis is provided

Blank space again

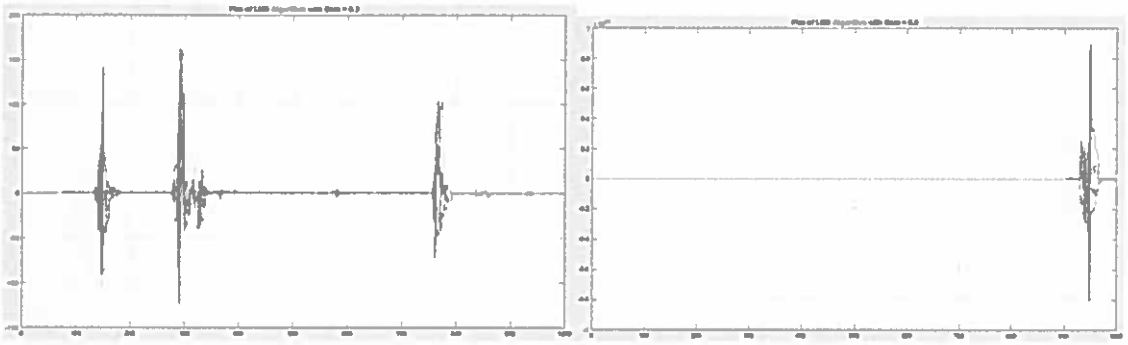
4.2 LMS algorithm

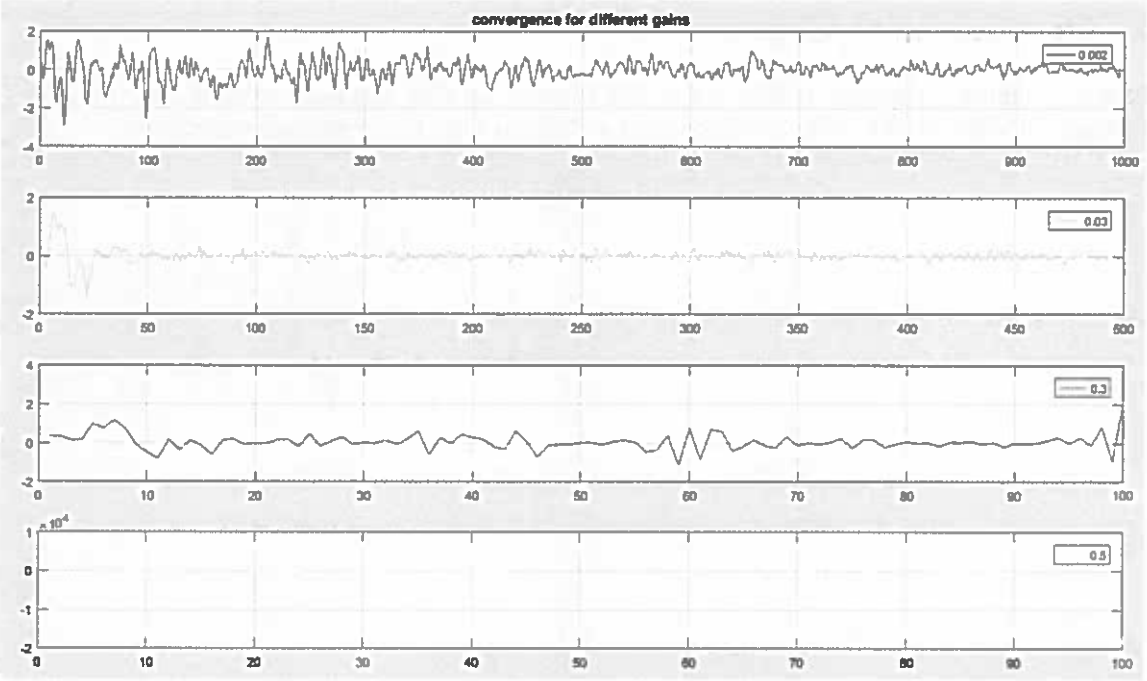
```
function [LMSest,Error,awevo ] = lms( x, z , u , O )
LMSest =[];
Error = [];
awevo = zeros(O, length(x)-O);
for n = O:length(x)
     LMSest(n,1)= awevo(:,n)'*x(n-O+1:1:n,1);
     xnorm = x(n-O+1:1:n,1);
     e = (z(n) - LMSest(n,1));
     Error = [Error e];
     aw = u*e*xnorm;
    awevo(:,n+1) = awevo(:,n) +aw;
end
end
```

Legend, title and axis are too small



Correct Plot of the evolution of the coefficients





From the plots above, higher gains reduces the time for convergence, thus increasing adaptation speed but introduces more errors and coefficient fluctuations.

Q3

For equation ue[n]x[n], n complexity.
For equation of error = z[n] y[n], n complexity.
For equation of w(n+1) = w(n) +ue[n]x[n], n^2 complexity.
Hence LMS has a total complexity of n^2.

Some complexixty analysis is provided

4.3 Gear shifting

Gear shifting enables the algorithm to change its learning rate, μ based on certain parameters. Large value of gain at start up is applied for fast convergence and then switching to a smaller value to improve the steady state response, reducing errors.

Not enough gear shifting ideas

4.4 Identification of AR processes

A1 and a2 should converge to -0.2870 and -0.9129 respectively according to my calculations in Matlab.
To study the effect of gain on the evolution of a1 and a2, the plots below have been generated:

Correct plot of the evolution of the AR coefficients

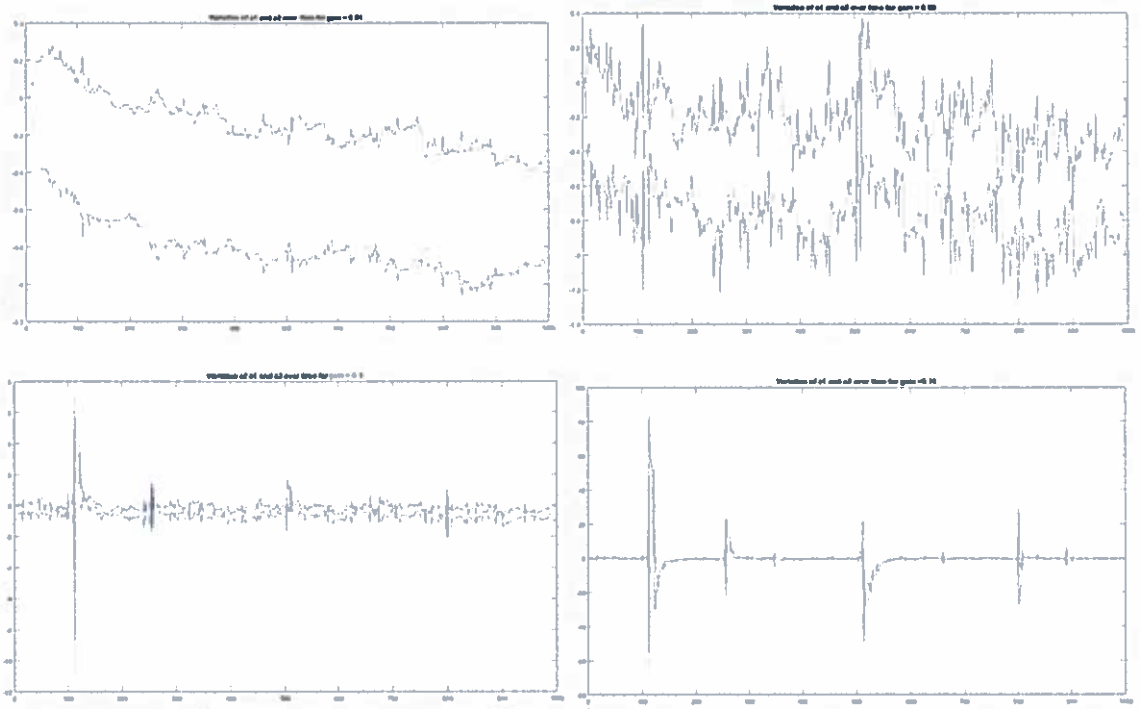


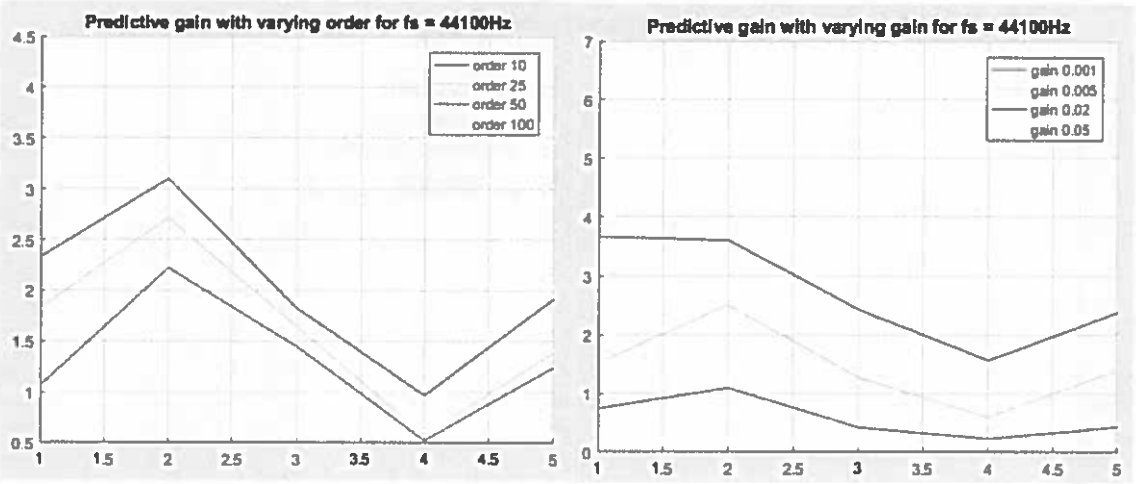
Figure 4.4 (a) Effect of varying gain on the evolution of a1 and a2.

For gain of 0.01, a1 and a2 converged to -0.2870 and -0.9129. No error.
For gain of 0.05, a1 and a2 converge to -0.2213 and -0.5747. Slight error but shorter time needed. And more fluctuations are introduced
For gain of 0.1, a1 and a2 converge to -0.1706 and -0.1647.
For gain of 0.15, a1 and a2 converge to -0.3760 and -0.8807.
Huge error and fluctuations even though convergence is almost instant.
We can see a trade-off between gain and error, as for small gains such as 0.01, it takes a long time to converge, but is very stable. For a high gain of 0.15, it converges almost instantly, but there is a high error present. Hence I would suggest using a gain of 0.05 or less to achieve more accuracy.

No explanation for negative values of AR coefficients is provided

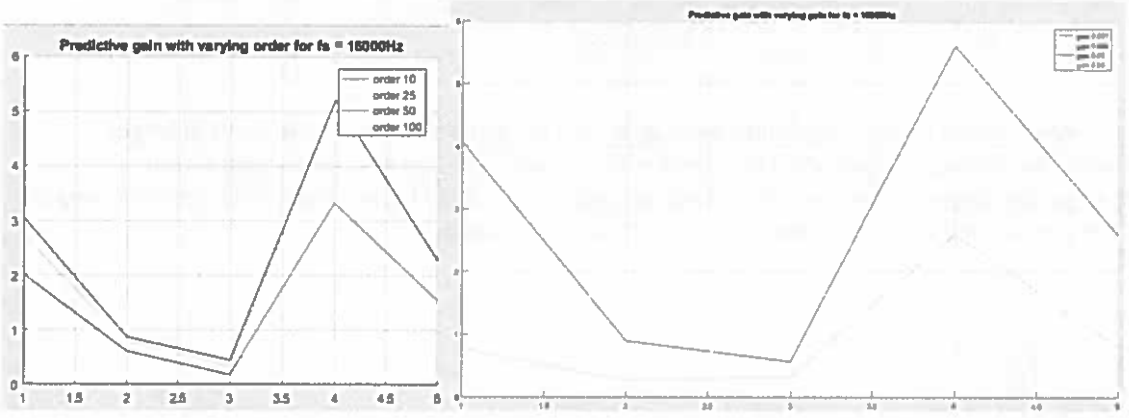
Missing plot of the time domain signals of the speech

4.5 Speech recognition



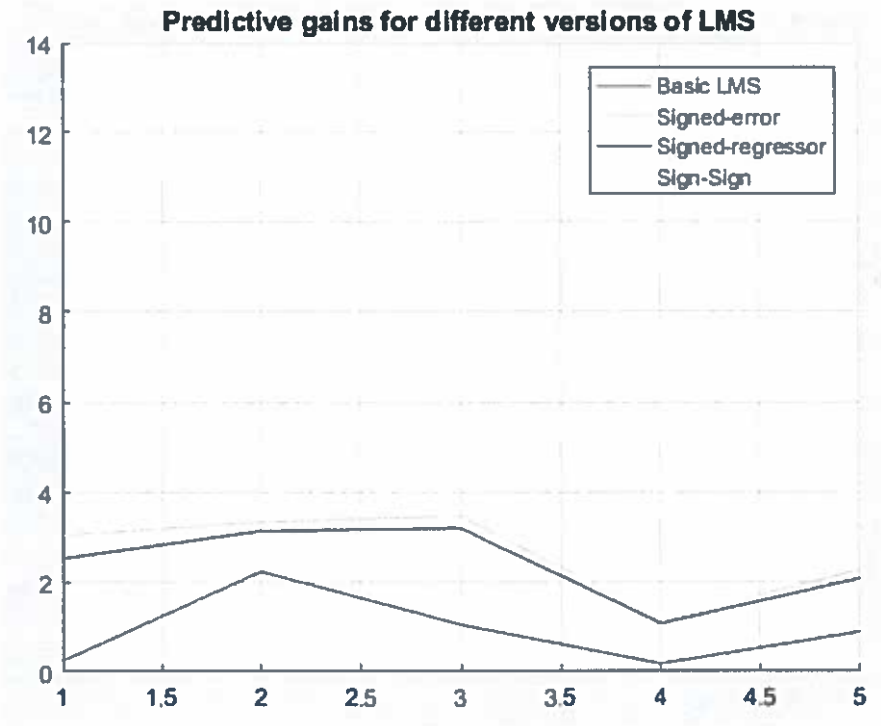
For a sampling frequency of 44100Hz, an increase in gain shows increase in predictive gain and decrease in errors. More orders also show better estimation and higher predictive gain.

A higher order and gain will give a high predictive gain at the cost of more complexity and speed.



Explanations are basically missing

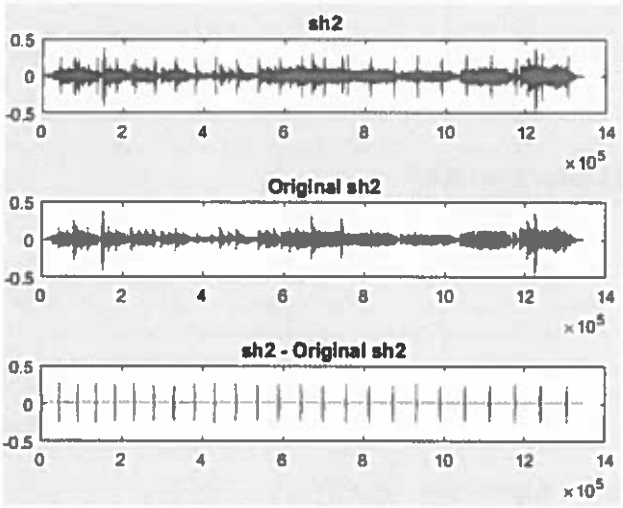
4.6 Dealing with computational complexity



The predictive gain for all 3 algorithms are higher than the basic algorithm, at the cost of longer convergence times and accuracy. The signed-regressor and sign-sign algorithms have slower convergence ranges but have smaller steady state error. The signed-error algorithm fastest converging speed but has the largest variance around the steady state value.

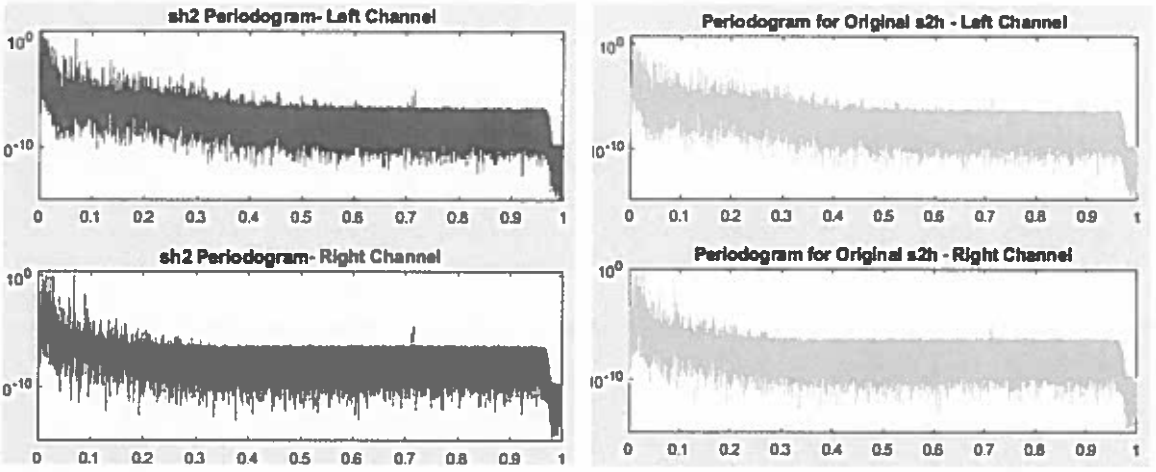
No complexity analysis is provided

5. A Real World Case Study: Vinyl Denoising



By taking the difference of s2h and its original track, it is evident that there exist a periodic signal(s) that corresponds to the whistling tickling. The signal(s) exhibit a certain frequency as seen qualitatively on the plot of s2h-Originals2h.

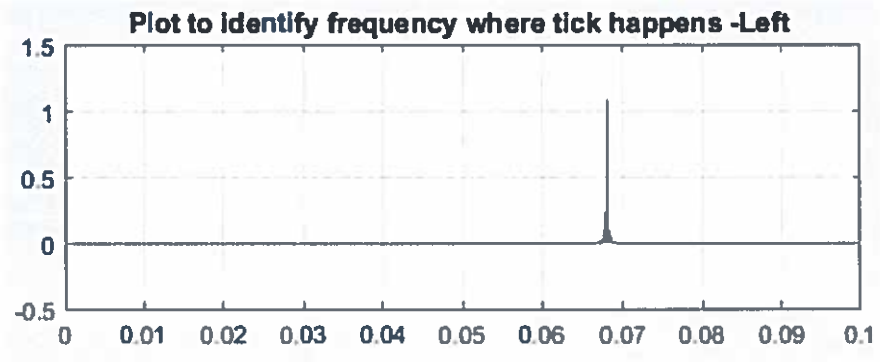
Figure 5.1 (a) Time series plot of s2h, original s2h and their difference



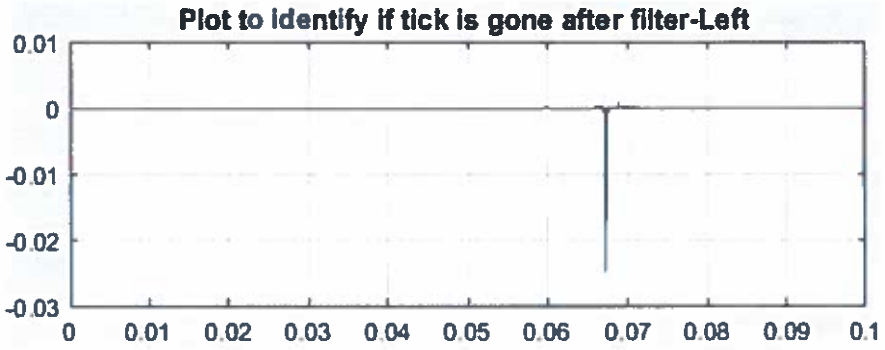
Caption and explanations are missing

Figure 5.1 (b)

By subtracting the original signal from the corrupted signal, the redundant frequency in the corrupted signal can be isolated out.

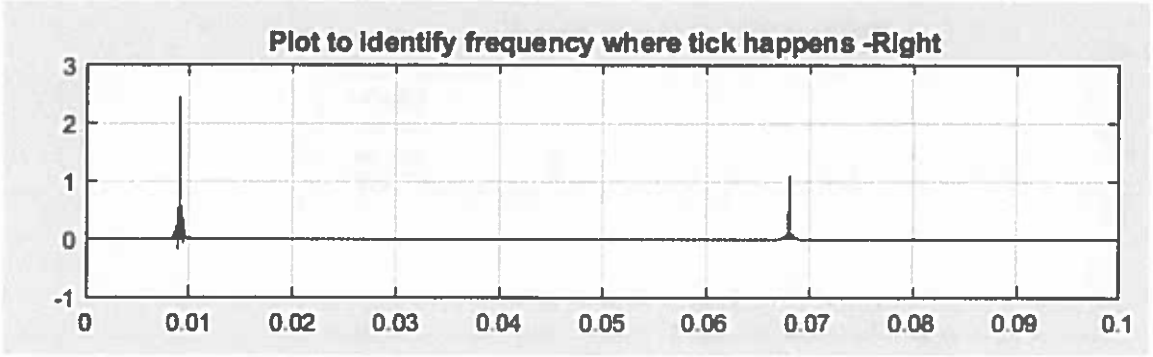


The tick has a frequency range of between 0.067 to 0.068.

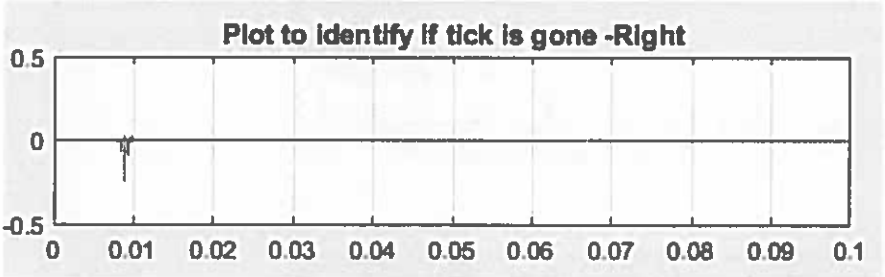


After applying a Butterworth filter with coefficient [0.0672 0.0688]
The amplitude of tick is reduced from 1.2 to -0.025.

No justification of the filter parameters



The tick frequency for right channel were found at 0.009 and 0.068.



After applying Butterworth filter with coefficient `[0.0674  0.0686] & [0.0088  0.0093]`
The amplitude of tick is reduced from 2.5 to -0.025 and the tick at frequency 0.068 it gone.

No computation of relative Error

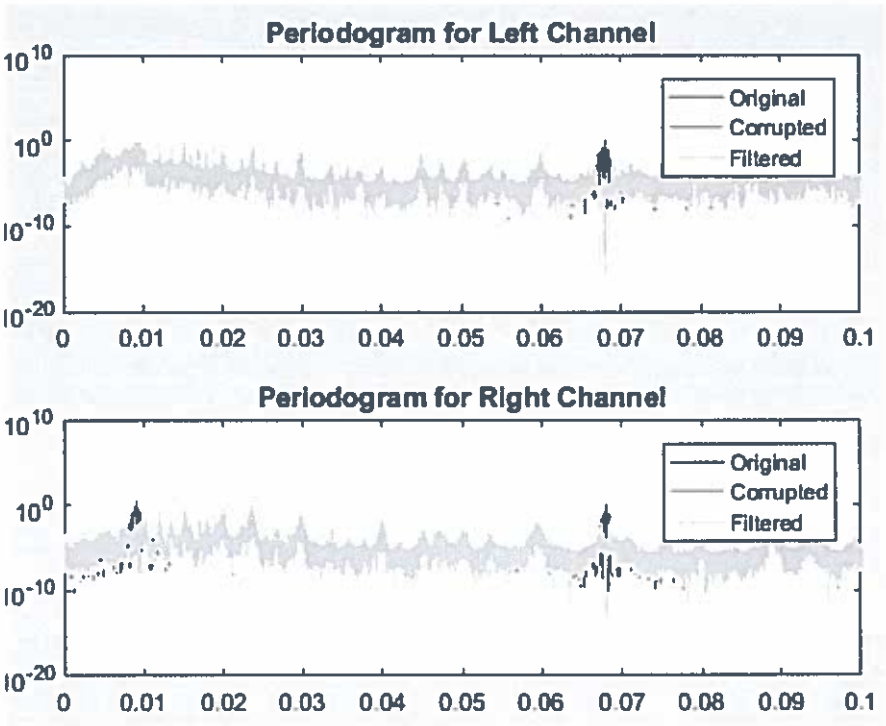


Figure 5.4(a) Periodogram of Original, Corrupted and filtered signals on same plot.

The relative errors for the left and right channel before filtering are 0.0247 and 1.147 respectively. After filtering the relative errors for left and right channel are 5.021e05 and 0.0354 respectively.
However, there is spectral component of the original song lost due to filtering, especially in the right channel as the spectral components of the song and the spectral component of the ticks are very near.

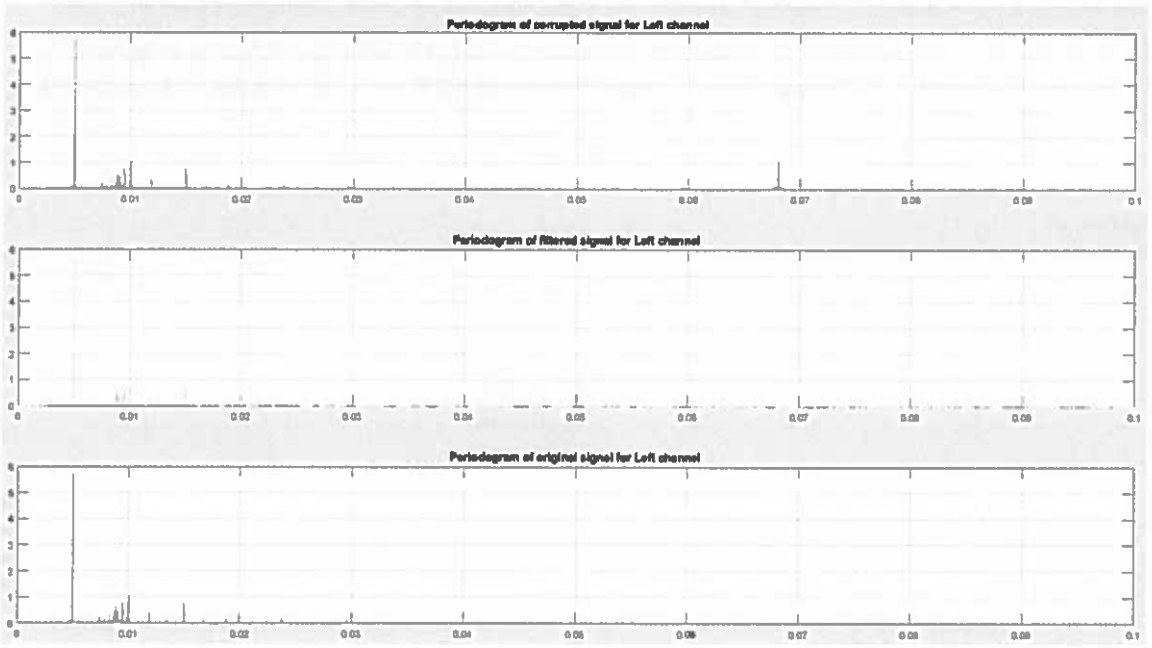
Q5.

NLMS algorithm

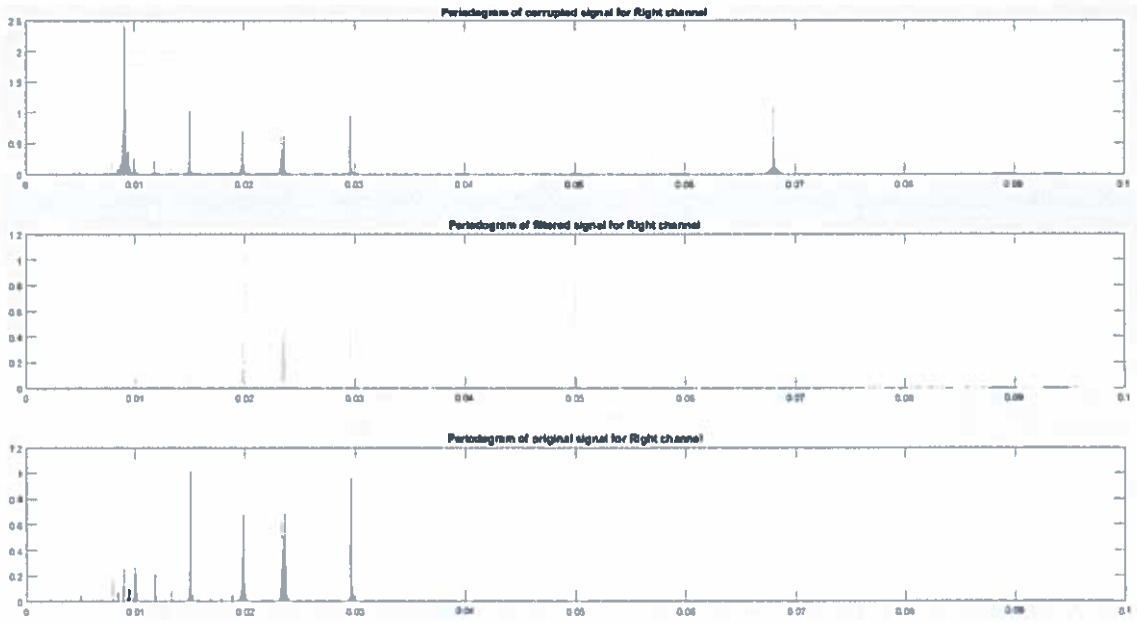
```
function [aw_evo,LMS,e] = NLMS(x,y_teach,order,gain)

aw_evo = zeros(order, length(x)- 1);
LMS =[];
e = [];
for n = order:length(x)
     LMS(n,1)= aw_evo(:,n)'*x(n-order+1:1:n,1);
     x_vector = x(n-order+1:1:n,1);
     norm_x = transpose(x_vector)*x_vector;
     error = y_teach(n) - LMS(n,1);
     e(n) = error;
     if norm_x == 0
     aw_evo(:,n+1) = aw_evo(:,n) + x_vector;
     else
     aw_evo(:,n+1) = aw_evo(:,n) + gain*error*x_vector/norm_x;
     end
end
```

Plotting the periodogram of the left channel using NLMS gives:



Plotting the periodogram of the right channel using NLMS gives:



For both plots, the spike due to the ticking at frequency 0.068 is attenuated.

We compare the cumulative squared error for the NLMS filter and the LMS filter and found that the error for the left and right channel using NLMS is 0.014 and 0.07 respectively and the error using LMS filter is 0..4004 and 0.1280 respectively. It is clear that the NLMS filter provides a better estimate than the LMS filter.

Some comparison of
the NLMS vs LMS

Part of the
assignment is
missing